# ClimateAdapt Pune: Evidence-Led Climate Action Report

Pilot Study: Pune Urban Core, Maharashtra | Generated: 18 October 2026 17:57 UTC

**Executive Summary:** This report translates real municipal drainage statistics (OpenCity Pune), OpenStreetMap infrastructure assets (5,779 trees, 267 medical facilities, waterways), Open-Meteo 7-day atmospheric forecasts, and Google Gemini AI into actionable climate resilience priorities for Pune's electoral wards. All findings distinguish measured infrastructure facts from model proxies, with automated MILP budget optimization and expert LLM strategic synthesis.

## 1. High-Priority Wards (Screening Indicators)

Scores are normalized 0-100 screening proxies. Higher values denote greater exposure or infrastructure deficit.

| Ward / Prabhag | Heat Proxy | Flood Proxy | Water Stress | Green Deficit | Composite |
|---|---|---|---|---|---|
| Vadgaon Sheri - Ganesh N | 58.4 | 75.2 | 52.0 | 95.0 | 68.8 |
| Arunkumar Vaidya Stadium | 64.4 | 54.8 | 60.0 | 95.0 | 65.0 |
| Ghorpade Garden | 63.8 | 54.6 | 60.0 | 93.3 | 64.4 |
| Upper Indiranagar | 58.4 | 59.1 | 60.0 | 95.0 | 64.4 |
| Mohammadwadi | 58.4 | 60.7 | 52.0 | 95.0 | 63.7 |
| Vimannagar Sanjay Park | 58.4 | 55.4 | 60.0 | 95.0 | 63.1 |

## 2. Prioritized Adaptation Action Plan (Budget Allocation)

**Target Portfolio Budget:** INR 5,000,000 | **Committed Allocation:** INR 5,000,000 (100.0% efficiency)

| Intervention | Category | Units | Cost (INR) | Dept | Lead Time |
|---|---|---|---|---|---|
| Shaded Bus Stops & Public Co | Heat & Urban Co | 12 station | ■2,160,000 | PMPML & PMC Health | 2 mo |
| Cool Roof Reflective Coating | Heat & Urban Co | 15 building | ■1,800,000 | PMC Building & Dev | 1 mo |
| Decentralized Rainwater Harv | Water Security | 2 building | ■440,000 | PMC Water Supply D | 2 mo |
| Critical Stormwater Drain De | Flood & Stormwa | 1 major dr | ■350,000 | PMC Drainage & Sew | 1 mo |
| Urban Shade Corridors & Nati | Heat & Urban Co | 1 km corri | ■250,000 | PMC Garden Departm | 3 mo |

## 3. Climate Stress Scenario Comparisons

| Scenario | Heat Index | Flood Index | Water Stress | Composite Index |
|---|---|---|---|---|
| Current Baseline | 58.4 | 75.2 | 52.0 | 68.8 |
| Extreme Heatwave (+3.5°C) | 78.4 | 75.2 | 67.0 | 78.0 |
| Extreme Monsoon Cloudburst | 48.4 | 100.0 | 42.0 | 76.0 |
| Post-Adaptation Plan | 39.6 | 75.2 | 49.8 | 61.6 |

## 4. Responsible Use & Data Quality Disclosures

• ***Distinction of Evidence:*** *Drainage statistics are verified 2015-2016 OpenCity PMC releases. Atmospheric values represent Open-Meteo multi-variable NWP forecast models and must not be cited as permanent climate normals.*
• ***Explicit Missing Data:*** *Local observation wells (GSDA telemetry) and uncalibrated DEM hydrological runoff grids remain unconfirmed and are strictly excluded from automated physical simulation to prevent false certainty.*
• ***Human Review:*** *Action allocations are exploratory decision-support calculations and require municipal site surveys before civil construction.*

# 5. AI Ward Resilience Brief: Ward #19 (Vadgaon Sheri - Ganesh Nagar)

**Analysis Engine:** Google models/gemini-3.8-flash | **Zone:** Nagar Road | **Evaluation Date:** 2026-10-18

## 1. What's Life Like in This Ward?

In Vadgaon Sheri - Ganesh Nagar, summer turns tin-roof homes into ovens with temperatures crossing 34°C and zero official trees to give shade. When monsoon arrives, life comes to a halt because 98% of our roads have no drains to carry rainwater away, causing immediate *pani bharaav*. On top of that, dirty water spills out into slum lanes where 8 out of 10 homes do not even have an underground sewer line.

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## 2. Problems and What We Can Do About Them

| Problem | How Bad Is It? | Why Does This Happen Here? | What Can Be Done? |
|---|---|---|---|
| **Street flooding (*pani bharaav*)** | 98.2% of our roads have no drains at all. | Rain has nowhere to go, so it fills up streets after every heavy shower. | Clean existing big *naalas* and put metal screens to stop trash from blocking water flow. |
| **Indoor heat** | Temperatures reach over 34°C, making homes unbearable. | Tin and concrete roofs soak up the sun and trap heat inside all day. | Apply white heat-reflecting paint on rooftops of 15 building clusters to keep rooms cooler. |
| **No shade on roads** | There are 0 recorded public trees and 0 parks in the ward. | Concrete roads and buildings have completely replaced open green spaces. | Put up shaded bus stops with cool drinking water and plant shade trees along a 1 km stretch. |
| **Overflowing dirty water** | 80% of slum homes lack underground sewer pipe connections. | Waste water runs into open street ditches that choke easily. | Clean the main ward drains urgently so dirty water does not mix with rainwater and back up. |
| **Wasting rainwater** | Rainwater is lost on roads while people face water shortages. | Rain simply washes away because buildings do not have storage systems. | Put rooftop rainwater collection tanks in 2 government office complexes to save clean water. |

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## 3. How Will the Money, Workers and Time Be Used?

| Work To Be Done | How Many? | Money (Rs.) | Workers Needed | Time Needed | Who Is Responsible? |
|---|---|---|---|---|---|
| Shaded bus stops with cold drinking water taps | 12 booths | Rs. 2,160,000 | 30 person-months | 2 months | PMPML & PMC Health Department |
| White heat-reflecting paint for rooftops | 15 building clusters | Rs. 1,800,000 | 30 person-months | 1 month | PMC Building & Development Department |
| Rooftop rainwater collection tanks | 2 municipal buildings | Rs. 440,000 | 6 person-months | 2 months | PMC Water Supply Department |
| Cleaning choked *naalas* and fitting trash filters | 1 major drain section | Rs. 350,000 | 5 person-months | 1 month | PMC Drainage & Sewerage Department |
| Planting local shade trees along the roadside | 1 km stretch | Rs. 250,000 | 4 person-months | 3 months | PMC Garden Department / Tree Authority |

• Total money being used:** Rs. 5,000,000 out of Rs. 5,000,000
• Money left over:** Rs. 0
• Workers needed:** 75.0 person-months
• Everything done in:** 3 months

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## 4. What Should the Ward Officer Do? (Step by Step)

1. **Week 1–2:** Clear out black mud and plastic trash from the main *naala* section and fit iron mesh screens so floating waste does not choke the water flow.
2. **Month 1:** Hire local painters to coat the roofs of 15 crowded building clusters with white heat-reflecting paint before peak heat hits.
3. **Month 2:** Set up the 12 shaded bus shelters with clean drinking water stations, and begin installing rooftop rainwater tanks on the 2 municipal office buildings.
4. **Month 3:** Finish planting sturdy local shade trees along the 1 km roadside stretch, install tree guards, and test all newly fitted rainwater tanks before the rains arrive.

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## 5. How Will Residents Know Things Are Getting Better?

- ■ After rain, streets should drain within 30-45 minutes instead of staying flooded for hours.
- ■ Homes with white-painted roofs should feel cooler in the afternoon, making rooms livable without fans blowing hot air.
- ■ Auto drivers, commuters, and street vendors can find cold drinking water and cool shade at 12 bus stops across the ward.